DRL-Based HVAC Control for a Single-Zone Smart Building: A Comparative Study of Agents

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***Abstract*—Buildings consume a large share of global energy, and HVAC is a major load. Efficient HVAC control is hard because weather, prices, and thermal dynamics vary over time. This paper studies deep reinforcement learning for continuous HVAC control in a single-zone smart building. We compare PPO, A2C, SAC, and TD3 in a physics-based environment with nonlinear heat transfer, a Carnot-based coefficient of performance, monthly electricity prices, domain randomization, and balanced seasonal sampling. Agents are trained on Syracuse weather and tested on held-out Albany weather to measure geographic generalization. Over 100 evaluation episodes, TD3 performs best with $0.12\,^{\circ}\mathrm{C}$ MAE, $16.9\,\mathrm{kWh/day}$ energy use, and 99.7% of time within $\pm 0.5\,^{\circ}\mathrm{C}$ of setpoint. A2C and SAC also generalize well, while PPO performs worse. These results show that a simple simulator can support strong performance on unseen weather when training includes seasonal diversity and parameter uncertainty.**

***Index Terms*—Deep reinforcement learning, HVAC control, smart buildings, energy efficiency, PPO, SAC, TD3, A2C, domain randomization**

I. Introduction

Buildings remain a major source of energy use, and heating, ventilation, and air-conditioning (HVAC) systems account for a large share of that demand. In practice, HVAC control is difficult because outdoor temperature, solar gains, wind speed, and electricity prices all change over time. Building envelopes also differ from one site to another, so a controller must make good decisions under uncertainty and over long time horizons.

Conventional approaches range from simple rule-based or proportional–integral–derivative (PID) controllers to model predictive control (MPC). MPC can plan ahead by optimizing over a building model, but it requires accurate calibration, forecasts, and nontrivial engineering effort, which limits scalability across buildings and climates. Deep reinforcement learning (DRL) offers a different path: a learned policy maps observations directly to control actions without solving a new optimization problem at each step. This is attractive for HVAC because the task is sequential, continuous, and affected by delayed consequences. However, generalization remains a central issue, since a controller that performs well in one climate or simulator may lose performance when the deployment context changes.

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This paper studies that question in a compact but informative setting. We use a single-zone thermal model with nonlinear conductance, solar gains, a coefficient-of-performance (COP)-aware energy term, and monthly electricity prices. We compare four continuous-control DRL agents: Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), Soft Actor-Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3). Training uses Syracuse weather only; evaluation uses Albany weather only, creating a strict geographic out-of-distribution test. We also add two training design choices aimed at improving generalization: domain randomization over thermal parameters and round-robin seasonal sampling so that each agent sees winter, spring, summer, and fall evenly during training.

This paper makes three contributions. First, it provides a controlled comparison of four DRL agents on the same single-zone HVAC task with identical observations, actions, and reward structure. Second, it studies generalization with a clear train–test split across cities, not just across episodes from the same data source. Third, it shows that a lightweight simulator with physics-aware reward design, domain randomization, and balanced seasonal exposure can produce policies that transfer well to unseen weather.

II. Related Work

Deep reinforcement learning has attracted growing interest for HVAC control because it handles sequential decision-making, continuous action spaces, and delayed consequences without requiring explicit building models. Recent surveys confirm that DRL methods now match or exceed MPC on comfort–energy trade-offs, while highlighting open challenges in reward design, reproducibility, and sensitivity to changing conditions [1]–[3].

On the algorithm side, the actor-critic family is especially relevant for continuous control. A2C/A3C introduced asynchronous advantage estimation, which improved training stability through parallel data collection [4]. PPO simplified trust-region policy updates with a clipped surrogate objective and became a widely used on-policy baseline [5]. For off-policy control, TD3 reduced value overestimation with clipped double critics, delayed policy updates, and target policy smoothing [6]. SAC added maximum-entropy learning

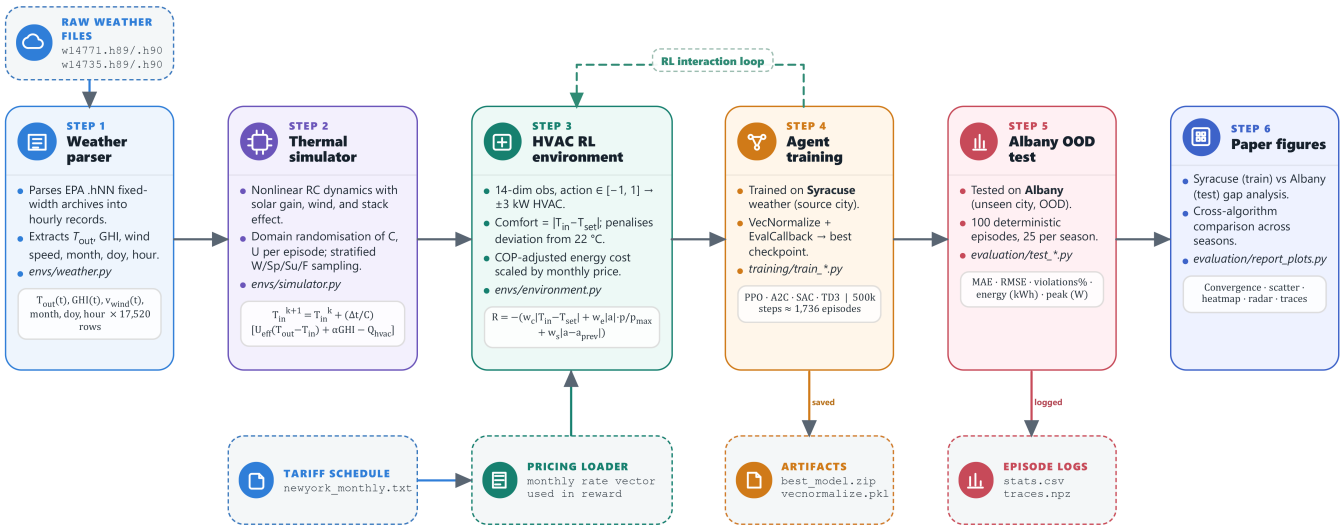

Fig. 1. EPA weather archives and New York electricity tariffs feed a nonlinear lumped-capacitance thermal simulator with domain randomization and stratified seasonal sampling. The simulator exposes a Gymnasium environment with a 14-dimensional observation space, a continuous HVAC power action, and a three-term reward balancing comfort, COP-adjusted energy cost, and actuator smoothness. Four DRL agents (PPO, A2C, SAC, TD3) are trained for 500 k steps on Syracuse weather and evaluated out-of-distribution on Albany over 100 deterministic episodes (25 per season). Resulting episode logs and trained checkpoints are post-processed into the comparative figures and tables of this paper.

with a stochastic actor and strong sample efficiency, making it a popular choice for stable continuous control [7].

Several recent studies compare these algorithms on HVAC tasks. The Sinergym benchmark [1] evaluated PPO, TD3, and SAC across different buildings and climates, finding that off-policy methods often achieve better comfort–energy trade-offs and arguing that many HVAC papers evaluate only one or two algorithms in a single narrow setting.

Generalization and transfer have emerged as a second major theme. A continual DRL study [2] showed that sample efficiency and transfer remain key bottlenecks; a model-based SAC framework with hypernetworks improved transfer across tasks and reduced catastrophic forgetting, suggesting that standard model-free training may not suffice when controllers must adapt across buildings. Along a complementary line, coupling DRL-based HVAC control with an urban climate model [3] demonstrated that background climate changes both the reward trade-off and the transferability of learned policies across cities.

Our work is closest in spirit to [1], but focuses on a different gap. We use a simpler single-zone setting so that the effect of algorithm choice is easier to isolate, and we test transfer with a strict geographic split between training and evaluation cities. We also combine domain randomization, balanced seasonal sampling, and a cost-aware reward that depends on both electricity price and instantaneous COP, bridging between high-level HVAC benchmarks and more controlled analysis of generalization.

III. System Model

Fig. 1 gives an overview of the end-to-end system pipeline, from raw data sources through the physics-based simulator and Gymnasium environment to training and evaluation.

A. Thermal Zone Dynamics

The indoor temperature evolves according to a first-law lumped-capacitance model, discretized with a forward-Euler step of $\Delta t = 300\,\mathrm{s}$:

$$T_{\text{in}}^{k+1} = T_{\text{in}}^{k} + \frac{\Delta t}{C}\left[U_{\text{eff}}\left(T_{\text{out}}^{k} - T_{\text{in}}^{k}\right) + \alpha \cdot \text{GHI}^{k} - Q_{\text{hvac}}^{k}\right] \quad (1)$$

where the effective envelope conductance captures conduction, wind-driven infiltration, and buoyancy-driven stack effects:

$$U_{\text{eff}} = U_{\text{cond}} + k_w\, v_{\text{wind}}^{n_w} + k_s\sqrt{|T_{\text{in}} - T_{\text{out}}|} \quad (2)$$

Table I lists the physical parameters and their domain-randomization ranges.

TABLE I
Thermal Simulator Parameters

Symbol	Description	Value / Range
C	Thermal capacitance	$\mathcal{U}(200\,\text{k}, 500\,\text{k})\,\text{J}/^\circ\text{C}$
U_{cond}	Conduction coefficient	$\mathcal{U}(30, 80)\,\text{W}/^\circ\text{C}$
k_w	Wind infiltration coeff.	$4.45\,\text{W}/^\circ\text{C}$
n_w	Wind power-law exponent	0.65
k_s	Stack effect coefficient	$2.6\,\text{W}/^\circ\text{C}$
α	Solar gain coefficient	$0.5\,\text{m}^2$
η_{COP}	Carnot efficiency factor	0.4
$Q_{\max}$	Max HVAC power	3000 W
T_{set}	Temperature setpoint	$22\,^\circ\text{C}$
Δt	Control timestep	300 s

B. Weather Data and Train/Test Split

We use US Environmental Protection Agency (EPA) PRZM-format hourly archives from 1989 to 1990 for two upstate New York stations:

TABLE II
14-DIMENSIONAL OBSERVATION SPACE.

#	Feature	Range	Group	Description
0	$T_{\text{in}} - T_{\text{set}}$	$[-50, 50]\,^{\circ}\text{C}$	Thermal	Indoor temperature error
1	$(T_{\text{out}} - 20)/20$	$[-2, 2]$	Thermal	Outdoor temperature, centered and scaled
2	$\sin(2\pi h/24)$	$[-1, 1]$	Time	Hour sine encoding
3	$\cos(2\pi h/24)$	$[-1, 1]$	Time	Hour cosine encoding
4	$\sin(2\pi d/365)$	$[-1, 1]$	Time	Day sine encoding
5	$\cos(2\pi d/365)$	$[-1, 1]$	Time	Day cosine encoding
6	$p_t/p_{\max}$	$[0, 1]$	Exogenous	Price normalized by annual maximum
7	GHI$/1200$	$[0, 1]$	Exogenous	Solar irradiance normalized by 1200
8	$v_{\text{wind}}/20$	$[0, 1]$	Exogenous	Wind speed normalized by 20
9	C/C_{nom}	$[0.57, 1.43]$	DR	Capacitance ratio
10	U/U_{nom}	$[0.55, 1.45]$	DR	Conductance ratio
11	a_{t-1}	$[-1, 1]$	Actuator	Previous action
12	COP$/10$	$[0.08, 1]$	Physics	COP normalized by 10
13	$U_{\text{eff}}/150$	$[0, 1]$	Physics	Effective conductance normalized by 150

- **Syracuse** (station 14771), used for training only.
- **Albany** (station 14735), used for evaluation only as the out-of-distribution (OOD) city.

Each archive provides about 17 520 hourly records of temperature, global horizontal irradiance (GHI), and wind speed. The two datasets have no overlapping records. Albany data is never used during training, tuning, or model selection. The split is geographic and enforced at the file-path level in the codebase, which prevents data leakage by construction.

C. Electricity Pricing

A 12-element monthly electricity-price vector based on 2025 New York residential rates is used to scale the energy term in the reward function. The same price schedule is shared across training and evaluation, since electricity pricing is a policy input rather than a weather-dependent feature.

IV. REINFORCEMENT LEARNING FORMULATION

A. Observation Space

The agent observes a 14-dimensional continuous vector at each timestep, summarized in Table II. Features are grouped into four categories: thermal state, time encodings, pricing and weather, and system parameters. Cyclic sine and cosine features encode hour of day and day of year without discontinuities. Exposing the randomized ratios C/C_{nom} and U/U_{nom} allows the policy to adapt online to different building envelopes.

B. Action Space

The agent outputs a single continuous action $a_t \in [-1,\, 1]$, mapped to HVAC power by $Q_{\text{hvac}} = a_t \times Q_{\max}$ (Table III). Positive values activate cooling. Negative values activate heating. The continuous formulation lets the agent modulate power smoothly instead of switching among discrete modes.

C. Reward Function

The agent receives a three-term penalty signal at each control step, designed to balance thermal comfort, energy efficiency, and actuator smoothness:

$$R_t = -\Big(\underbrace{w_c\,|T_{\text{in}} - T_{\text{set}}|}_{\text{comfort}} + \underbrace{w_e\,\frac{|a_t|}{\text{COP}_t}\cdot\frac{p_t}{p_{\max}}}_{\text{energy cost}} + \underbrace{w_s\,|a_t - a_{t-1}|}_{\text{slew penalty}}\Big) \tag{3}$$

The comfort term penalises temperature error directly. The energy term divides action magnitude by the instantaneous COP, so the same control effort costs more when the system is less efficient, and then scales it by time-of-use price. The slew penalty discourages abrupt changes in HVAC power. The default weights are $w_c = 1.0$, $w_e = 0.1$, and $w_s = 0.05$. This makes comfort the main objective while keeping the controller cost-aware and smooth.

D. Domain Randomization

At each episode reset the thermal capacitance C and conduction coefficient U_{cond} are drawn uniformly from the ranges in Table I. The agent observes the ratios C/C_{nom} and U/U_{nom}, which allows it to adapt to unknown building envelopes at test time.

E. Stratified Seasonal Sampling

Naive random episode sampling from a two-year weather archive can over-represent mild periods and under-sample the

TABLE III
ACTION SPACE

Property	Value
Dimension	1 (Box)
Range	$a_t \in [-1,\, 1]$
Mapping	$Q_{\text{hvac}} = a_t \cdot Q_{\max}$
Max. power	$Q_{\max} = 3000\,\text{W}$
$a_t > 0$	Cooling
$a_t < 0$	Heating
$a_t = 0$	HVAC off

hardest heating and cooling conditions. We therefore partition all valid 24-hour episode start indices into four buckets: winter, spring, summer, and fall. At each reset, the simulator cycles through these buckets in round-robin order and draws a random start index from the active bucket.

The same protocol is used during evaluation. The 100 test episodes contain exactly 25 episodes from each season. This keeps the reported metrics from being dominated by whichever season contributes the most raw data.

V. Algorithms

We compare four continuous-action DRL algorithms implemented in Stable-Baselines3 [8]: two on-policy methods, PPO [5] and A2C [4], and two off-policy methods, SAC [7] and TD3 [6].

A. Proximal Policy Optimization (PPO)

PPO is an on-policy policy-gradient method that updates the actor with a clipped surrogate objective. The clipping term limits how far the new policy can move from the old policy in one update, which usually improves stability. In our setting, PPO is a strong baseline because it is simple and widely used, but its on-policy nature can make it less sample efficient than off-policy methods.

B. Advantage Actor-Critic (A2C)

A2C belongs to the actor-critic family introduced through asynchronous training by Mnih et al. It learns a policy and a value baseline together, using advantage estimates to reduce gradient variance. Compared with PPO, A2C is lighter and often faster to train, though it can be more sensitive to hyperparameters and rollout settings.

C. Soft Actor-Critic (SAC)

SAC is an off-policy actor-critic method based on the maximum-entropy objective. It learns a stochastic policy that seeks high reward while also keeping enough entropy for exploration. Replay-buffer reuse and entropy regularization make SAC an attractive choice for HVAC control, where data efficiency and stable learning are both important.

D. Twin Delayed Deep Deterministic Policy Gradient (TD3)

TD3 is an off-policy deterministic actor-critic method designed to reduce value overestimation in continuous control. It uses two critics, delayed policy updates, and target policy smoothing to stabilize learning. These features are well suited to precise HVAC actuation, where small value errors can lead to poor power commands.

VI. Experimental Setup

A. Training Configuration

All agents are trained for 500 000 environment steps on Syracuse weather data. On-policy methods, PPO and A2C, use 8 parallel `SubprocVecEnv` workers. Off-policy methods, SAC and TD3, use a single `DummyVecEnv`. Observations are normalized via `VecNormalize`. Reward normalization is applied only for on-policy agents. Table IV lists the key hyperparameters.

TABLE IV
Training Hyperparameters

Parameter	PPO	A2C	SAC	TD3
Total steps		500 000		
Policy		MlpPolicy [64, 64]		
Seed		42		
# envs	8	8	1	1
Learning rate	3e-4	7e-4	3e-4	3e-4
γ	0.98	0.99	0.98	0.98
Batch size	64	–	256	256
n_{steps}	2048	5	–	–
Buffer size	–	–	1 M	1 M
τ	–	–	0.02	0.005
Entropy coeff.	0.01	0.01	auto	–
Action noise	–	–	–	$\mathcal{N}(0, 0.1)$
Reward norm.	yes	yes	no	no

B. Evaluation Protocol

Each trained agent is evaluated deterministically for 100 episodes on the unseen Albany weather data, with 25 episodes from each season. We report the median of five metrics: mean absolute error (MAE), root-mean-square error (RMSE), time within $\pm 0.5\,^{\circ}\mathrm{C}$ of setpoint, daily energy consumption, and peak HVAC power.

VII. Results

A. Learning Curves

Fig. 2 shows the mean evaluation reward during training. TD3 and SAC converge smoothly and plateau by 400k steps. A2C is competitive despite early instability. PPO reaches a lower reward plateau, consistent with its weaker test-time performance.

B. Generalization Performance

Table V summarizes the overall Albany test-set performance. TD3 achieves the lowest MAE ($0.12\,^{\circ}\mathrm{C}$) and the lowest energy use ($16.9\,\mathrm{kWh/day}$) while maintaining near-perfect comfort with 99.7% of time within $\pm 0.5\,^{\circ}\mathrm{C}$ of setpoint. A2C matches TD3 on comfort but consumes slightly more energy. PPO is the weakest method on all main metrics and reaches the comfort band only 64.9% of the time.

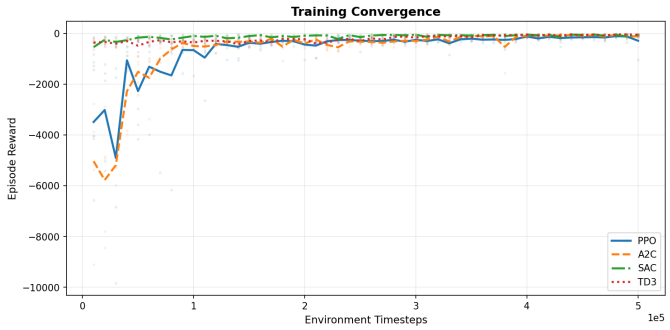

Fig. 2. Mean evaluation reward during 500k-step training with interquartile range shading.

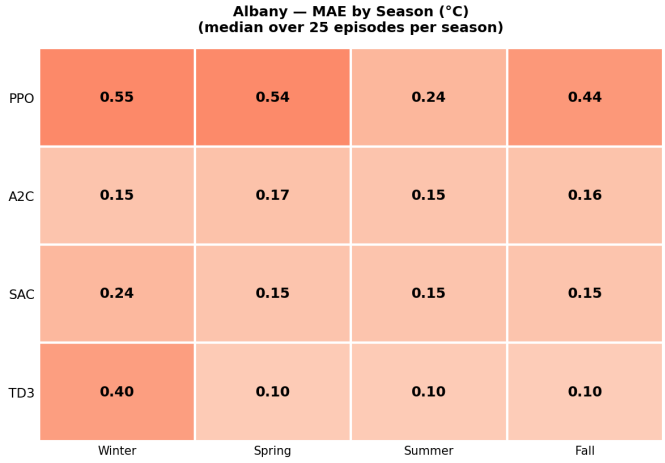

Fig. 3. Median MAE (°C) on Albany (out-of-distribution) over 25 episodes per season. Lower is better.

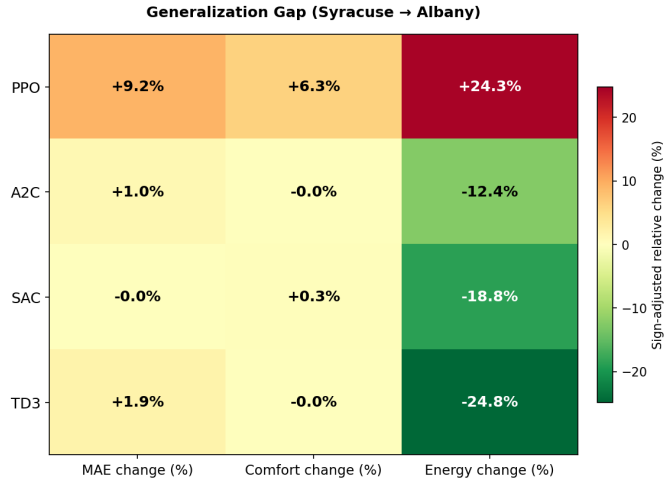

Fig. 4. Percentage change in median performance when transferring from Syracuse (training) to Albany (test). Positive values indicate worse transfer performance on Albany; comfort is reported as loss in percentage points.

Here, comfort refers to the percentage of time the indoor temperature stays within $\pm 0.5\,^{\circ}\mathrm{C}$ of the desired setpoint; higher values mean the controller keeps the building closer to its target temperature. Comparing Albany (test) with Syracuse (train) shows that the policies transfer well. A2C, SAC, and TD3 maintain nearly identical tracking accuracy on Albany: median MAE changes by at most $0.02\,^{\circ}\mathrm{C}$, and comfort remains above 99%. Energy consumption actually *decreases* on Albany for these three agents (e.g. TD3 drops from $22.5\,\mathrm{kWh/day}$ on Syracuse to $16.9\,\mathrm{kWh/day}$ on Albany), which reflects Albany's milder climate rather than a policy deficiency. PPO is the exception: its MAE rises from 0.41 to $0.45\,^{\circ}\mathrm{C}$ and its comfort drops below 65%, indicating that PPO did not learn a policy robust enough to transfer across cities under the current training budget.

TABLE V
MEDIAN ALBANY PERFORMANCE (100 OOD EPISODES)

Metric	A2C	PPO	SAC	TD3
MAE (°C)	0.16	0.45	0.17	**0.12**
RMSE (°C)	0.19	0.50	0.22	**0.15**
Within ±0.5 °C (%)	**99.7**	64.9	99.0	**99.7**
Energy (kWh/day)	18.5	21.7	21.0	**16.9**
Peak Power (W)	2657	3000	**2415**	2397

Fig. 3 breaks performance down by season. All algorithms struggle most in winter, where the larger indoor to outdoor temperature difference increases both tracking difficulty and energy use. TD3 and A2C keep at least 99.7% comfort across all four seasons, while PPO drops sharply in spring.

Fig. 4 shows how performance changes when moving from the Syracuse training distribution to the unseen Albany test distribution. Comfort degradation is very small for A2C, SAC, and TD3, while PPO shows a much larger drop. All agents also use less energy on Albany, which is consistent with a milder test climate.

C. Seasonal Analysis

Fig. 5 shows representative winter and summer episodes for TD3, the best-performing algorithm overall. These seasons were chosen because they reflect the most demanding heating and cooling conditions. In winter, the controller applies sustained heating of about $2000\,\mathrm{W}$ to counter a $-10\,^{\circ}\mathrm{C}$ outdoor temperature while keeping the indoor temperature inside the comfort band. In summer, less HVAC effort is needed and the policy changes power smoothly.

D. Discussion

Three patterns stand out. First, TD3 is the best overall method in this setting. Its deterministic policy appear to fit a low-dimensional continuous control problem where precise power levels matter. The gains are visible in both comfort and energy use.

Second, A2C is a stronger baseline than expected. Although it is simpler than SAC and TD3, it reaches the same comfort rate as TD3 and stays close on tracking error.

Third, transfer performance seems to depend not only on the algorithm but also on the training setup. Domain randomization exposes agents to different thermal masses and conductances, while stratified seasonal sampling reduces overfitting to mild weather. Although these experiments do not separate the effect of each choice, the small comfort gap for A2C, SAC, and TD3 on the unseen city suggests that both help generalization.

VIII. CONCLUSION

This paper presented a controlled comparison of four DRL agents—PPO, A2C, SAC, and TD3—for continuous single-zone HVAC control under a common nonlinear thermal simulator, reward function, and evaluation protocol. All agents were trained exclusively on Syracuse weather and evaluated on 100 deterministic episodes of unseen Albany weather (25

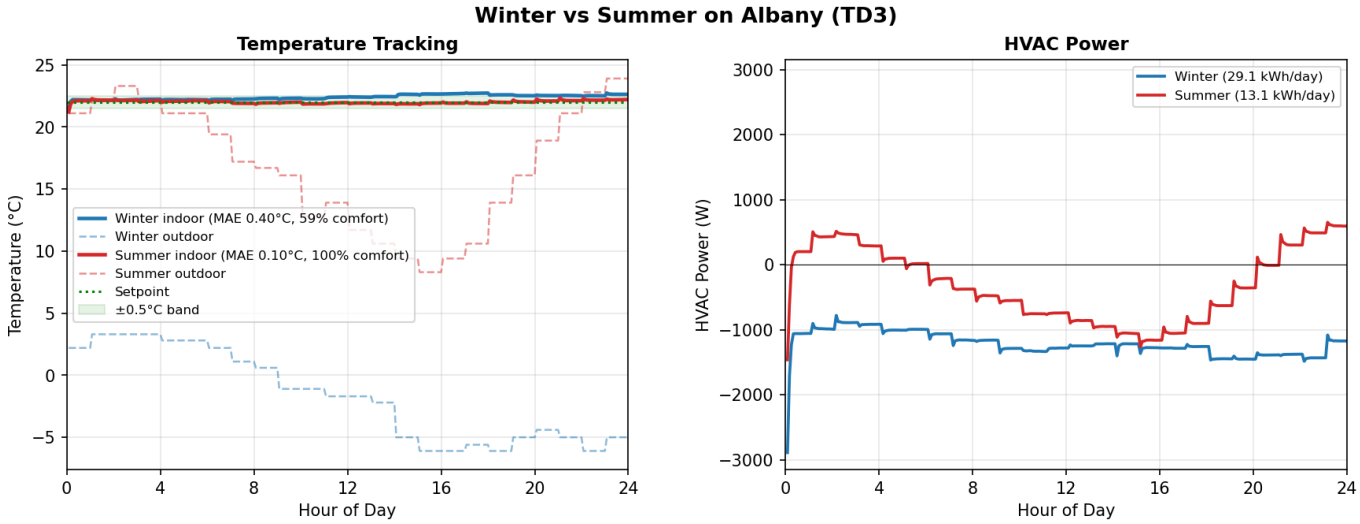

Fig. 5. Representative winter and summer Albany episodes for TD3. Left: indoor and outdoor temperature traces with the $\pm 0.5\,^{\circ}\mathrm{C}$ comfort band in green. Right: HVAC power profile for each season.

per season), providing a strict geographic out-of-distribution test.

TD3 achieved the strongest overall performance on the Albany test set, recording a median MAE of $0.12\,^{\circ}\mathrm{C}$, an RMSE of $0.15\,^{\circ}\mathrm{C}$, 99.7% of time within $\pm 0.5\,^{\circ}\mathrm{C}$ of the $22\,^{\circ}\mathrm{C}$ setpoint, and daily energy use of only $16.9\,\mathrm{kWh/day}$ with a peak draw below $2400\,\mathrm{W}$. PPO was the weakest agent, with an MAE of $0.45\,^{\circ}\mathrm{C}$, only 64.9% comfort, the highest energy consumption at $21.7\,\mathrm{kWh/day}$, and actuator saturation at the $3000\,\mathrm{W}$ system limit.

The generalization gap analysis confirmed that A2C, SAC, and TD3 all transferred well from Syracuse to Albany, with comfort losses below one percentage point. All agents used less energy on Albany than on Syracuse, consistent with Albany's milder climate. PPO was the outlier, showing a substantially larger comfort drop on the unseen city.

Two training-design choices contributed to these results. Domain randomization over thermal capacitance C and conductance U forced each agent to handle a range of building envelopes, while round-robin seasonal sampling ensured balanced exposure to winter heating and summer cooling extremes. Although the current experiments do not isolate the individual contribution of each mechanism, the small generalization gap for three of four agents supports their combined effectiveness.

Several directions remain open. Extending the simulator to multi-zone buildings with inter-zone heat flow would test scalability. Adding occupancy schedules, demand-response events, and time-varying tariffs would increase realism. Formal ablation studies on domain randomization and seasonal sampling are needed to quantify their separate effects. Finally, increasing the training budget beyond $500\,\mathrm{k}$ steps or tuning PPO-specific hyperparameters may close the gap between PPO and the stronger agents.

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